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Assembly manipulation of small objects by dual-arm manipulator

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Abstract

Purpose – The purpose of this paper is to propose an assembly robot that exhibits specific human-like skills, with minimal structural cost and a number of external sensors.

Design/methodology/approach – The authors have employed vision processing using multiple cameras to determine targets and postures and propose strategies to determine the pose of a target and to prevent collisions between the fingers and obstacles in an environment with mixed objects. Furthermore, a dynamic trajectory planner integrates the vision and force sensors of the robot hand for the assigned task.

Findings – The authors obtained satisfactory experimental results for autonomous real-time grasping and screwing. The results verified the capability of the robot for handling small objects.

Research limitations/implications – More effective robotic manipulation requires a higher degree of target orientation data, which will be a future study of this research.

Practical implications – Practicality has been established through results, indicating the capability of the robot to implement human-like skilled manipulation of small objects. This can potentially reduce the high labor cost associated with the small-scale manufacture of custom-made products.

Originality/value – Screwing of nuts of minimum M2 size (diameter, 4.6 mm) and M8-M10 bolts (head diameter, 15-19.6 mm; length, 50-80 mm) by cooperating two seven-link arm manipulators and three-fingered hands shows the robot's capability to manipulate small objects.

Keywords Dual-arm robot, Image processing, Object recognition, Collision avoidance, Assembly

Paper type Research paper

Introduction

Vision sensors have emerged as important sensing elements for anthropomorphic multi-arm robots that manipulate objects. Vision-processing techniques have attracted considerable attention in recent studies on object recognition and pose calculation. Scale invariant feature transform (SIFT) (David, 2004) is one example, which detects image features to compute feasible grasp points (Collet *et al.*, 2009; Morales *et al.*, 2006). Other examples include a learning algorithm that detects the grasp points of an unknown object on the basis of multiple 2D images (Saxena *et al.*, 2007, 2008a, b). However, these methods require a large database for computation, and do not focus on assembly parts, but on common household objects, which do not require accurate detection for manipulation.

Interest-point detector studies for object recognition include work by Lowe (2004) and Leonardis *et al.* (2006). Lowe proposed the SIFT technique, and Leonard is proposed the speeded up robust features technique. They have shown that their feature vectors do not vary when subjected to object translation, scaling, and rotation, which provides a robust six

degrees of freedom (DOF) object pose calculation. However, their method does not perform well with featureless objects that are particularly small in size and without an obvious color, such as those used in assembly works. Another example of object posture calculation is the ARToolkit algorithm for augmented reality applications, proposed by Kato and Mark (1999) and Kato *et al.* (2000). However, this algorithm requires a large and clear border line pattern or marker, which makes the detection rigid and inapplicable for small objects.

Low-cost robots that exhibit human-like skills have attracted considerable attention owing to a need for autonomous small-scale manufacturing of specialized products. Several studies in related areas have revealed the capability of vision sensors for robot tasking. Dual-arm robots manufactured by FANUC Ltd (2010) and YASKAWA Electric Corporation (2003-2010) employ lasers and cameras for autonomous operation. Other examples include a vision-guided servo system on the PUMA robot for tracking a moving carburetor gasket (Feddema and Mitchell, 1989), and a hybrid application of eye-in-hand (EIH) and eye-to-hand (ETH) multi-cameras on the Comau SMART-3 S robot (Lippiello *et al.*, 2007). However, these robots manipulate objects that are bigger than assembly parts, such as nuts and bolts, which require accurate manipulation. Therefore, we have designed our dual-arm manipulator to do complicated tasks that involve small objects as small as M8 bolts (head diameter, 15 mm; length, 50 mm) and M2 nuts (diameter, 4.6 mm). Initially, we assign the robot the task of jointly manipulating objects for an autonomous screwing task.

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The proposed vision system calculates the posture of the target and integrates with a force sensor on the robot hand in order to obtain successful results.

Dual-arm manipulator system

Figure 1 shows the dual-arm manipulator 3, which has seven DOF and a three-fingered hand. A stereo camera is fixed on top of the robot as the ETH camera, and two monocular cameras are mounted on the end effector (EEs) as the EIH cameras. The left hand is equipped with six-axis force sensors at the finger tips, and the right hand is equipped with strain gauge sensors at the finger joints, in order to reduce the cost. The communication network is developed on the basis of the user datagram protocol and transmission control protocol of the internet protocol suite. Data socket functions are used for receiving and transmitting data between the main PC and the subsystems in a MATLAB environment. The structure of the robot is shown in Figure 2, with the specifications listed in Table I.

Camera to robot transformation

Perspective projection and reconstruction of 3D data from 2D stereo vision

Using a physical pinhole camera model, let $Q = (X_c, Y_c, Z_c)$ be a point on a distant object, projected onto an image plane at point $q = (u, v, f)$. Then, the relation between Q and q is expressed by equation (1). Here, $f = (f_x, f_y)$ is the focal length in the x - y directions, c_x and c_y denote the displacement of the principal point (the point of intersection of the center of the image plane and the optical axis), and s is the scaling factor that corresponds to the effective pixel size in meters:

$$\begin{bmatrix} X_c \\ Y_c \\ Z_c \end{bmatrix} = s \begin{bmatrix} f_x & 0 & c_x \\ 0 & f_y & c_y \\ 0 & 0 & 1 \end{bmatrix}^{-1} \begin{bmatrix} u \\ v \\ 1 \end{bmatrix} \quad (1)$$

Obtaining the 3D data using a stereo camera is useful for the coordinate system of the robot. However, target recognition using depth imaging is difficult, because of the small dimensions of the objects on the worktable. Therefore, we have reconstructed Z_c using the triangulation method (Bradski and Kaehler, 2008).

Robot space transformation

First, we can transform the camera coordinate system to the robot coordinate system (X_r, Y_r, Z_r) for the ETH camera data using equation (2) and Figure 3(a). The camera is rotated about the normal to the worktable by θ_c , and it then calculates the center of the object in the x - y directions (the X_c and X_r axes are perpendicular to this paper). Let r_{CaPa} denote the position of the object in camera coordinates according to equation (1). Then, we can calculate the position vector, r_{APa} , for moving the EEs toward the object as follows:

$$r_{APa} = Rot_x(\theta_c) r_{CaPa},$$

$$\text{where } Rot_x(\theta_c) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos \theta_c & -\sin \theta_c \\ 0 & \sin \theta_c & \cos \theta_c \end{bmatrix}. \quad (2)$$

Second, we can calculate r_{APa} , the position vector of the EE for grasping an object on the basis of the EIH camera data, using equation (3) and Figure 3(b). In equation (3), r_{ACa} denotes the direct kinematics calculation for the robot links and camera in robot coordinates. Furthermore, $Rot_z(\theta_c)$ and C_{AH} denote the rotation matrix for the camera and robot links, respectively, (similarly, both the camera axis, X_c , and robot axis, X_r , are perpendicular to this paper):

$$r_{APa} = r_{ACa} + C_{AH} Rot_z(\theta_c) r_{CaPa},$$

$$\text{where } Rot_z(\theta_c) = \begin{bmatrix} \cos \theta_c & -\sin \theta_c & 0 \\ \sin \theta_c & \cos \theta_c & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad (3)$$

In equation (3), r_{ACa} denotes the position of the EIH camera in robot space calculated using the following direct kinematics equation, based on the left-arm model in Figure 4. The position matrix, r , and the orientation matrices of the links, C , are expressed by equations (5)–(7), in which θ_1 – θ_7 are the rotation angles throughout the links. The EIH camera position relative to the seventh link is denoted by W in the model of the arm, and it is denoted by r_{CM} in the xyz direction:

$$r_{ACa} = r_{AP} + C_{AB} r_{PQ} + C_{AC} r_{QR} + C_{AD} r_{RS} + C_{AE} r_{ST} \\ + C_{AF} r_{TU} + C_{AG} r_{UV} + C_{AH} r_{CM} \quad (4)$$

Figure 1 Dual-arm manipulator 3

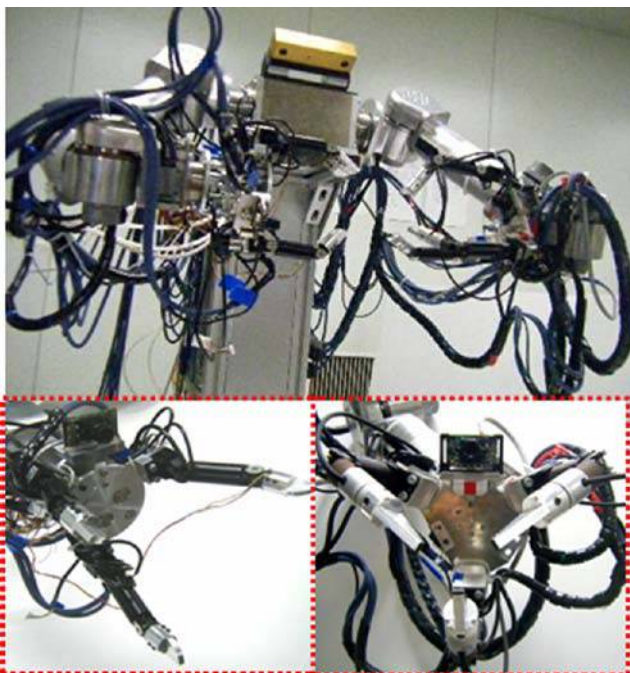
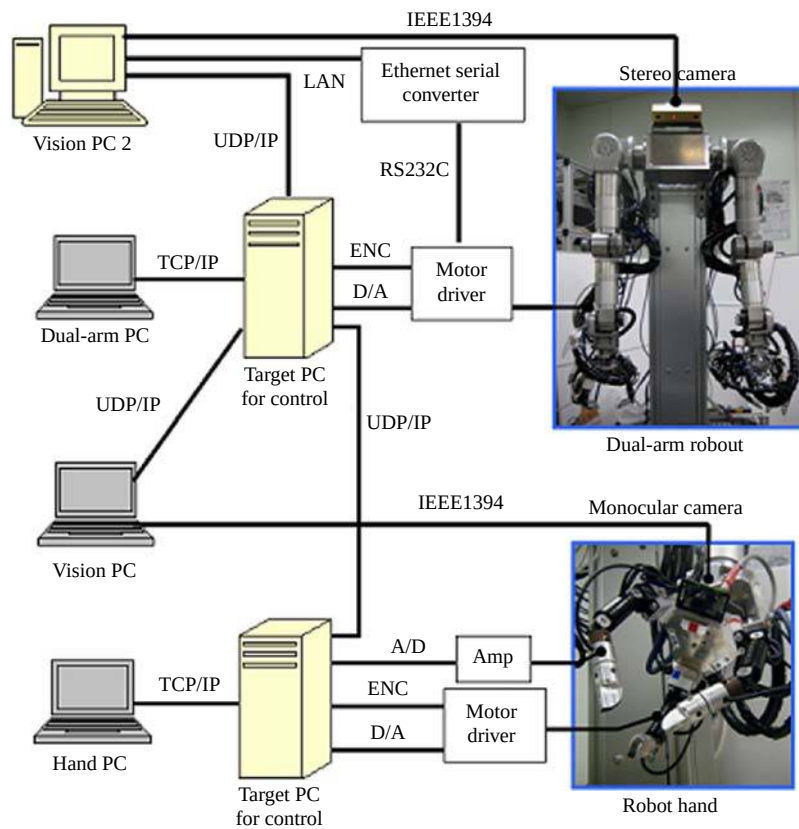


Figure 2 Robot system structure

Table I Dual-arm manipulator 3 specifications

Item	Robot arm	Robot hand
Size (L × W × H)	0.9 × 0.7 × 1.6 m	0.15 × 0.16 × 0.20 m
Weight	50 kgf	1.2 kgf
Degree of freedom	7-DOF × 2 arms	2-DOF × 3 + 1-DOF
Actuator	AC servomotors	AC servomotors
Articulation sensor	Absolute and incremental encoder	Incremental encoder
External sensor	Stereo camera and monocular camera	Six-axis force sensors × 3 (left hand) Strain gauge × 6 (right hand)
EES	Three-finger robot hand (2-DOF × 3) Three-finger robot hand (2-DOF × 3, 1-DOF rotation)	
Payload	3 kg/arm	

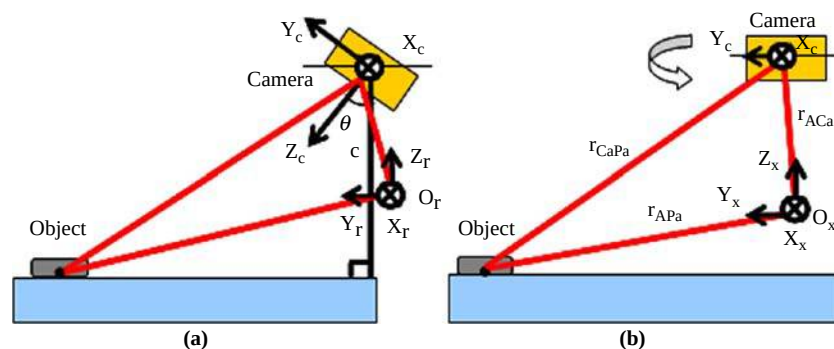
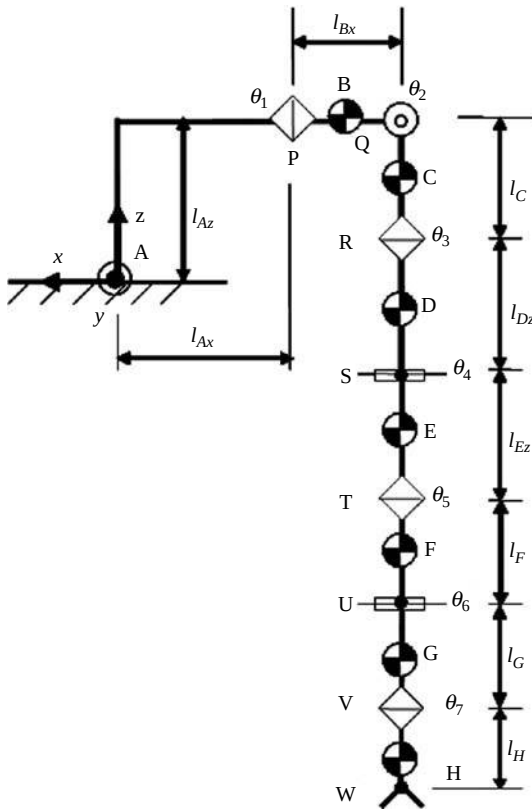
Figure 3 ETH and EIH camera coordinates


Figure 4 Left-arm model



$$\begin{aligned}
 r_{AP} &= \begin{bmatrix} -l_{Ax} \\ 0 \\ l_{Az} \end{bmatrix}, r_{PQ} = \begin{bmatrix} -l_{Bx} \\ l_{By} \\ 0 \end{bmatrix}, r_{QR} = \begin{bmatrix} 0 \\ 0 \\ -l_C \end{bmatrix}, \\
 r_{RS} &= \begin{bmatrix} 0 \\ l_{Dy} \\ -l_{Dz} \end{bmatrix}, r_{ST} = \begin{bmatrix} 0 \\ -l_{Ey} \\ -l_{Ez} \end{bmatrix}, r_{TU} = \begin{bmatrix} 0 \\ 0 \\ -l_F \end{bmatrix}, \\
 r_{UV} &= \begin{bmatrix} 0 \\ 0 \\ -l_G \end{bmatrix}, r_{CM} = \begin{bmatrix} l_{Hx} \\ l_{Hy} \\ -l_{Hz} \end{bmatrix}
 \end{aligned} \quad (5)$$

$$\begin{aligned}
 C_{AC} &= C_{AB}C_{BC} \\
 C_{AD} &= C_{AB}C_{BC}C_{CD} \\
 C_{AE} &= C_{AB}C_{BC}C_{CD}C_{DE} \\
 C_{AF} &= C_{AB}C_{BC}C_{CD}C_{DE}C_{EF} \\
 C_{AG} &= C_{AB}C_{BC}C_{CD}C_{DE}C_{EF}C_{FG} \\
 C_{AH} &= C_{AB}C_{BC}C_{CD}C_{DE}C_{EF}C_{FG}C_{GH}
 \end{aligned} \quad (6)$$

$$\begin{aligned}
 C_{AB} &= Rot_x(\theta_1), C_{BC} = Rot_y(\theta_2), C_{CD} = Rot_z(\theta_3), \\
 C_{DE} &= Rot_x(\theta_4), C_{EF} = Rot_z(\theta_5), C_{FG} = Rot_x(\theta_6), \\
 C_{GH} &= Rot_z(\theta_7)
 \end{aligned} \quad (7)$$

Image processing method

In this section, we discuss our method that is the result of modifying several available image processing algorithms in order to solve the constraints placed on the system by the small, featureless target objects that do not have an obvious color.

Importance of the geometric method for object recognition

We have used a geometric feature extraction method for recognizing and calculating the target in order to ensure accuracy via the direct calculation of pixel values. For comparison purposes, we also tested a learning algorithm that uses a Haar-like feature extraction method (Lienhart and Maydt, 2002) for object detection. This learning algorithm was effective when detecting objects and it was robust in an unstructured environment. However, there were a large number of false detection and position errors. Hence, it is unsafe and impractical for the autonomous manipulation of small M2-M10 nuts with diameters of 4.6–19.6 mm (Figure 5). In comparison, the geometric method yielded reliable results and accurate calculations for the targets of interest.

Recognition for the grasping task

The vision processing architecture is shown in Figure 6. We discuss the collision avoidance method in the next section.

Nut recognition begins by removing unrelated color pixels from the image with the background removal method. Here, we obtain a binarized image with nut-like objects as white pixel regions and mark them using the labeling algorithm. In the first step, the recognition step, we apply Sklansky's

Figure 5 Hexagonal nuts

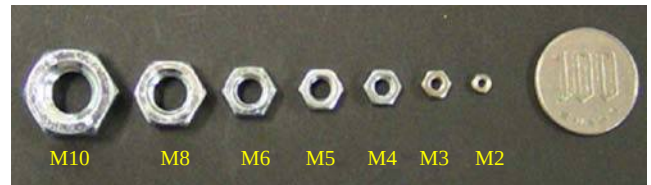
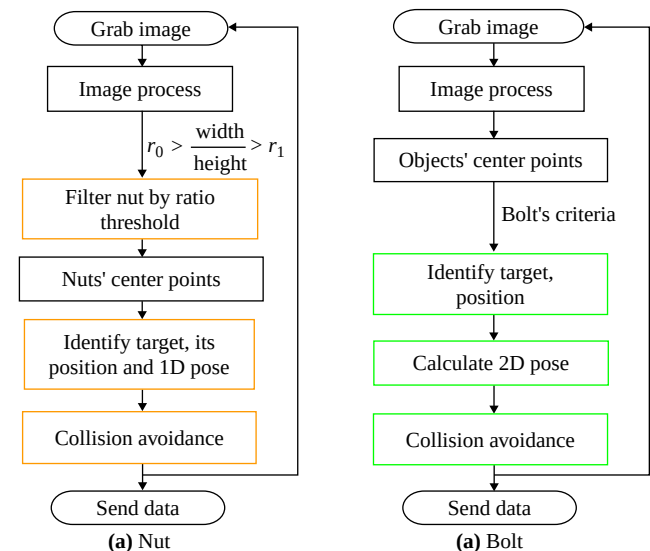


Figure 6 Image processing



algorithm (Bradski and Kaehler, 2008) to obtain the hull points around the object. However, this is not sufficient to detect the correct vertices of nut as the interest points for nut recognition. Thus, the detected hull points are turned into vectors and filtered through a scalar vector calculation, assuming that a hexagonal nut has a fixed range relationship between the vectors around each vertex. The steps are as follows:

- 1 For $i = 0, 1, 2, \dots, p$ and $k = 1, 2, \dots, n$, let the detected hull points and vertex points be denoted as pt_i and $vertex_k$.
- 2 Initially, pt_0 is set as the first vertex point, $vertex_1$. We use $vertex_1$ and pt_1 to create vector M_1 and compare it to the second vector, M_2 , which is created from the successive pt_i points.
- 3 Calculate $dot = M_1 \cdot M_2$ of a hexagonal nut as the threshold. As i increases, if $M_1 \cdot M_2 < |dot|$, then increase k . If a vertex is found at point $i = m$, update vector M_1 using pt_m and pt_{m+1} .
- 4 If $n = 6$, the object is recognized as a nut, otherwise exit and capture next object.

Consequently, by taking advantage of the symmetrical property of nuts of any size, we can apply the minimal area rectangle method (Bradski and Kaehler, 2008) to calculate the size and center point of the nut, as shown by the blue outline in Figure 7(a). Next, we need to select the largest nut for the grasping strategy. Here, we assume that choosing the largest nut first will decrease possible collisions between the fingers and the surrounding objects. All the nuts are compared, and when the largest nut is identified, we calculate its pose in the x - y directions (grasping angle, θ) using equation (8). The points (x_1, y_1) and (x_o, y_o) refer to the rightmost vertex and the center of the target, respectively. The point (x_1, y_1) can be identified from the rightmost detected vertex points (Figure 7(a)), and (x_o, y_o) is the center of the enclosed rectangle. Here, obtaining the correct value of θ ensures a secure grasp of the target by the robot hand. We also need to calculate the size of the target in order to implement different modes for grasping different size nuts:

$$\theta = \arctan\left(\frac{y_o - y_1}{x_1 - x_o}\right) * \frac{180}{\pi} \quad (8)$$

Next, we recognize a bolt based on the total number of corners (green dots) identified from the hull points (orange-colored line) (Figure 7(b)) surrounding the bolt. In the following steps, for $i = 0, 1, 2, \dots, n$, (x_i, y_i) denotes the detected 2D hull points, and the green dots are labeled counter-clockwise from the top left as p_1 - p_4 :

- Let $j, k \in i$. If $dist_x = x_{(i+1)} - x_i$ or $dist_y = y_{(i+1)} - y_i$ satisfies $\beta_1 < dist_x, dist_y < \beta_2$ at j and k , then, $p_1 = (x_j, y_j)$; $p_2 = (x_{j+1}, y_{j+1})$, $p_3 = (x_k, y_k)$, and $p_4 = (x_{k+1}, y_{k+1})$. Here, $dist_x, dist_y$, and β are the predefined thresholds.
- If $\sum p = 4$, the object is recognized as a bolt.
- We can use the distances between p_1 and p_4 to identify bolt sizes, and to differentiate between the head and tail of a bolt. In addition, we use these values to generate an orthogonal line across both ends to calculate the center of the bolt.
- Calculate the head and tail position from c . Use them to calculate the yaw angle (covered in the next section) and roll angle (θ_r) of the bolt. Here, θ_r is the angle of the orthogonal line in Figure 7(c), and is calculated using equation (8). We also need to convert the angle from image to camera coordinates.

Finally, repeat the vision algorithm described above in order to recognize multiple objects, as follows:

- Apply the background removal method to the captured image.
- For $i = 0, 1, 2, 3, \dots, k$, label the objects as $O_{m1}, O_{m2}, O_{m3}, \dots, O_{mk}$, respectively.
- Filter unnecessary blots or noise to reduce computation.
- Obtain the hull points for O_{m1} and execute the algorithm to recognize the targets. Calculate all the centers, including those of non-target objects.
- Repeat b . until the k th object. Identify the largest target from d . and label it as O_{max} . Treat the other objects as obstacles.
- Calculate the pose angle of O_{max} for grasping purposes. Use the position information of all the objects to apply the collision avoidance method.

Template matching for the screwing task

Template matching is a technique that identifies small parts of an input image that match a given template image. Here, we apply the normalized correlation coefficient matching method (Rodgers and Nicewander, 1988) because it has the advantage of reducing the lighting differences between the two images. The purpose is to align both the EEs in a screwing posture, as shown in Figure 8. The center of the tail of the bolt is detected by matching a template image over the image viewed from the left camera.

Firm grasp solution

For an autonomous screwing task, it is important to grasp nuts and bolts with maximum contact to prevent slippage. If grasping is impeded by nearby objects, the contact point

Figure 7 Target characteristics (top view)

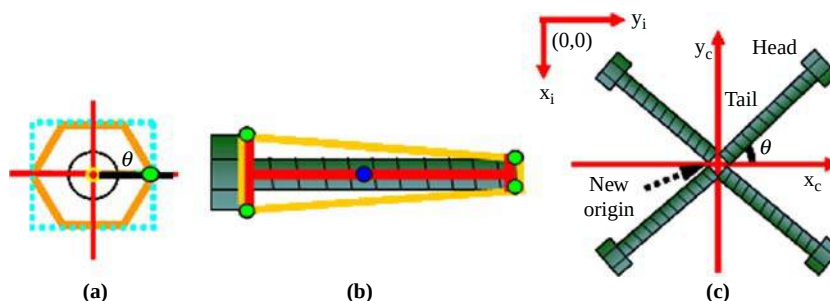
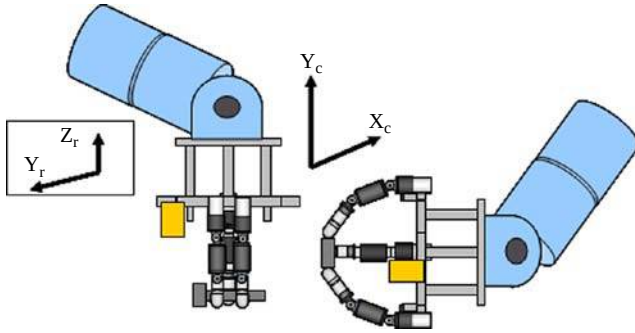


Figure 8 Matching nut to center of bolt



provided by vision will be affected. In the case of a bolt, as shown in Figure 7(b), the arm needs to change its posture according to the pose of the bolt to ensure that the fingers hold the bolt horizontally prior to the screwing task, thus eliminating vision errors before the alignment step. As a result, we propose the following methods to achieve this.

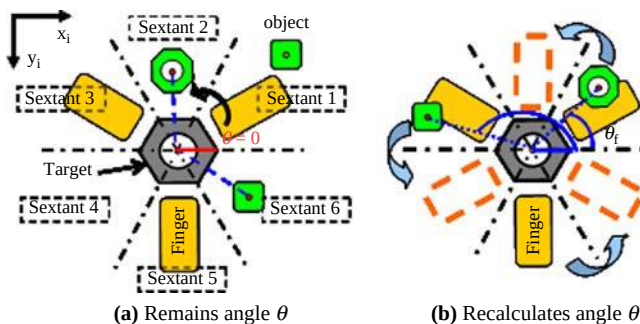
Occlusion detection for nut grasping

An iterative loop determines both the distance between the target and any nearby object(s), and the existence of nearby objects in odd or even sextants. The former is calculated using the Pythagorean theorem (equation (9)), where (x_{o1}, y_{o1}) and (x_o, y_o) are the central points of the nearby and the target objects, respectively. We determine the existence of objects by calculating the nearby object angle, θ_t (equation (10)), where $i = 1, 2, 3$. If the nearby objects are not in the same sextants as the initial contact points of the three fingers, i.e. in sextant 1, 3, and 5, as shown in Figure 9(a), then θ will remain as calculated by equation (8). Otherwise, the fingers need to avoid being blocked, as shown in Figure 9(b). Hence, equation (11) is used in both cases, and if the fingers are blocked in both regions, they will not grasp the nut. Positive value of θ is shown by the arrow in Figure 9(a):

$$r = \sqrt{(x_{o1} - x_o)^2 + (y_{o1} - y_o)^2} \quad (9)$$

$$\theta_t = \begin{cases} \text{odd sextants} & ; \quad 60(2)(i-1) + \theta < \theta_t < 60(2i-1) + \theta \\ \text{even sextants} & ; \quad 60(2i-1) + \theta < \theta_t < 60(2i) + \theta \end{cases} \quad (10)$$

Figure 9 Nut grasping angle



$$\theta = \begin{cases} \theta & ; \quad \theta_t \text{ even} \\ \theta + 60^\circ & ; \quad \theta_t \text{ odd \& } \theta < 0 \\ \theta - 60^\circ & ; \quad \theta_t \text{ odd \& } \theta > 0 \\ \text{no grasping} & ; \quad \theta_t \text{ even \& odd} \end{cases} \quad (11)$$

The same method is applied to calculate the distance of the bolt from other objects, using equation (9). However, owing to the comparably small size of a bolt, and to ensure that the fingers grasp the body of the bolt orthogonally, which is important for the screwing task, the method identifies only the distance, as shown in Figure 10.

Correcting arm pose for bolt grasping

Figure 11 shows the required pose of the robot arm (shown by a solid arrow) for it to grasp the center of the bolt. Let a , b , and d be the parameters calculated by vision. Then, we can calculate θ_p , the bolt's yaw angle, as follows:

$$c = \frac{a-d}{2} \quad (12)$$

and:

$$\theta_p = \arctan\left(\frac{c}{b}\right) \quad (13)$$

In robot coordinates, θ_p denotes the rotation of the object about the x -axis associated with the seventh link. Thus, from equation (3), we obtain equation (14), which we use to generate the trajectory of the robot autonomously:

$$r_{APa} = r_{ACa} + C_{AH} \text{Rot}_x(\theta_p) \text{Rot}_z(\theta_c) r_{CaPa} \quad (14)$$

Trajectory planning and strategies

We have designed a trajectory planner to enable us to adopt strategies for handling data from multiple sensors in order to

Figure 10 Bolt grasping angle (top view)

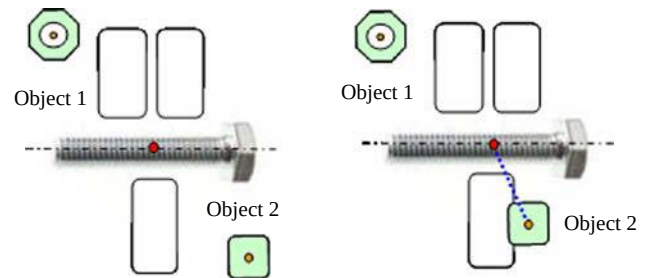
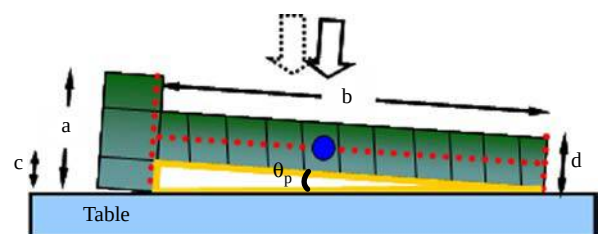


Figure 11 Yaw angle of bolt (side view)



generate a dynamic trajectory in real-time and in unknown situations. The strategy we use to implement the two tasks, i.e. autonomously grasping and then screwing in the nut and bolt in a continuous operation, is shown in Figure 12. Vision is the first stage. Here, we capture the target information via ETH and EIH cameras. The posture of the target is determined according to the previous strategy, prior to grasping. The force sensor of the left hand is then used to measure the moment and force of contact with the target or environment to compensate for small position errors (Saiki *et al.*, 2010). Furthermore, with respect to the screwing strategy, after making the first contact between both targets, the robot is assigned to re-check the force measurement and correct the position of the nut by itself, until it finds the best fit and begins to rotate the EE.

Experiment and results

A worktable with a black surface is placed in front of the robot. Normal lighting from ceiling fluorescent lamps is used throughout the experiment. We used M2-M10 nuts and M8 bolts (length, 50 mm) for the grasping test, and M10 nuts and bolts (length, 80 mm) for the screwing task. We execute PI position control on a windows operating system at a speed of 200 Hz, while vision processing is performed on a separate system with a processing speed that corresponds to the camera frame rate. ETH vision calculates the 3D object position within approximately 1 s. EIH vision needs a minimum of 136 ms for recognizing a single object, and a minimum of 200 ms to completely execute the vision algorithm for multiple objects.

Nut and bolt shape recognition

Figure 13 shows that the ETH algorithm identifies a small object inside the rectangular region and calculates the center of the object (marked with a red point). Then, as the arm approaches the object, the recognition begins. The results from the monocular camera in Figure 14 shows that the

proposed vision has performed well when detecting multiple objects and recognizing various size nuts (M8, M4 and M2) and bolts (M8 and M10).

Task 1: grasping the nut and bolt

An online experiment has been implemented to validate the grasping performance based on vision data. Only the positions in the x -direction are shown in the following results. Let the actual position of the EE and reference of the x -position be denoted by r_x and ref_x , respectively. In addition, the x -positions from ETH and EIH are denoted by r_{bbx} and r_{ffx} , respectively.

First, for bolt detection (Figure 15(a)), we used r_{bbx} , labeled “vision data” at 517 s, to calculate the position trajectory, ref_x , labeled “EE position” at 519 s. Consequently, in Figure 15(b), the EE moved according to the trajectory ref_x (at 525 s), generated according to r_{ffx} (at 523 s), to position r_x . A similar result is observed for nut detection, as shown in Figure 16. Hence, these examples show the effectiveness of the vision algorithm and generated trajectory when grasping an object.

Consequently, grasping tests were carried out for an M8 bolt and for M8, M4, and M2 nuts. The objective was to pick a nut and bolt in two test modes. In the first test, for the sake of repeatability, the robot had to pick the nut and bolt from the same point. In the second test, the robot needed to pick the nut and bolt from random points on the worktable. In each case, there were ten trials. For the bolt, the robot achieved 100 percent success when grasping, in both tests. Similarly, for a nut, the results in Table II shows that a satisfactory success rate was achieved. The results also show that the robot successfully integrated the vision and force sensor data when grasping small objects. However, position errors caused by vision and insufficient arm control made it difficult to grasp millimeter-scale target objects. As a result, further enhancement is required.

Next, for the collision avoidance test of the nuts, we performed ten real-time experiments by placing obstacles in

Figure 12 Operation flow of robot task

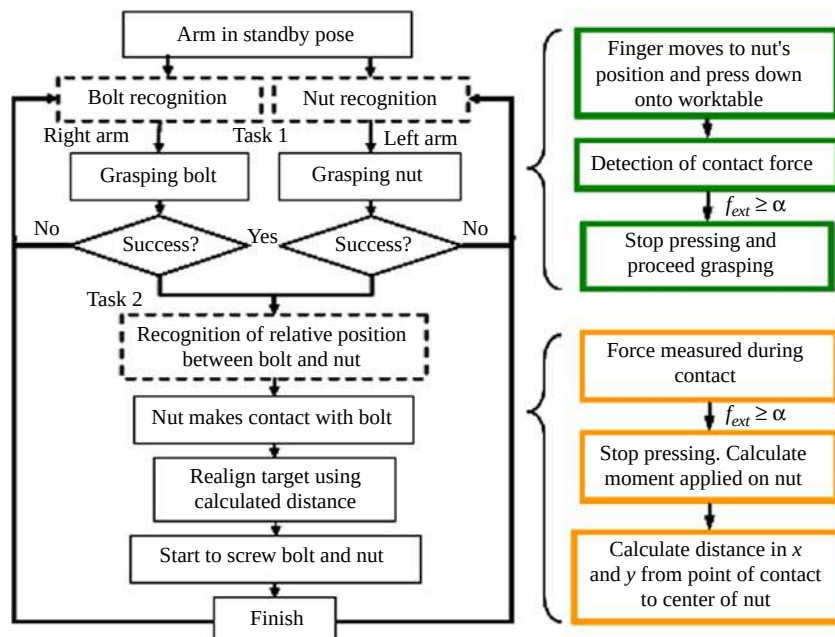


Figure 13 Detection by ETH camera

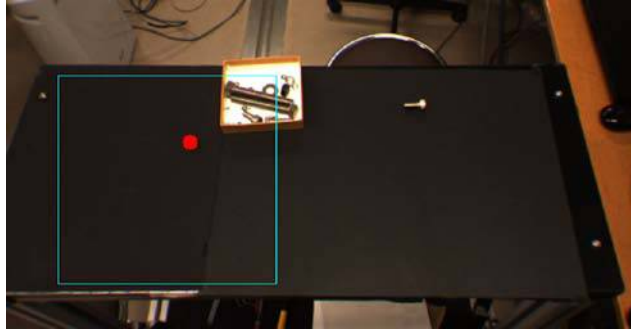


Figure 14 Recognition by EIH camera

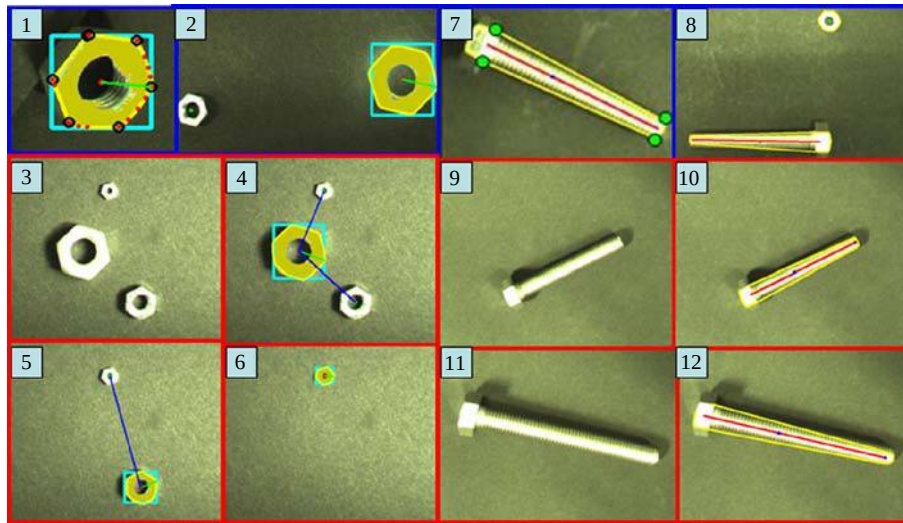
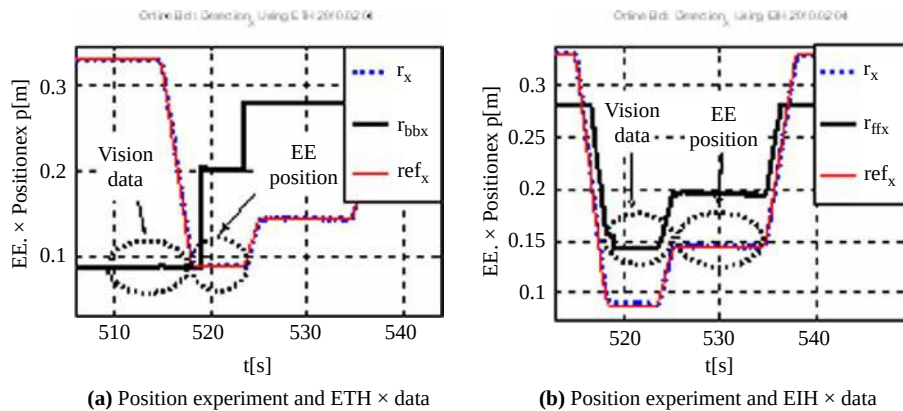


Figure 15 x-position for bolt grasping



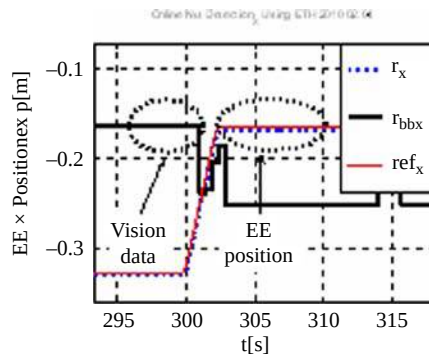
odd and even sextants around the target. The success rate is in Table III, and the EIH camera images are shown in Figure 17. The green and blue lines in Figure 17(a) and (b) show the detected locations of obstacles identified in odd and even sextants, respectively. In addition, the effect on the seventh link angle in Figure 18 shows that the vision data, $\Theta_{7_{vision}}$, affected the current seventh joint angle, $\Theta_{7_{before}}$, preventing the fingers being blocked, as shown by $\Theta_{7_{Left}}$ around 470 and 520 s.

In the case of the bolt, the distance between the bolt and the nearby object could be clearly identified in Figure 17(c) and (d). Overall, the method for collision avoidance has shown successful results when grasping.

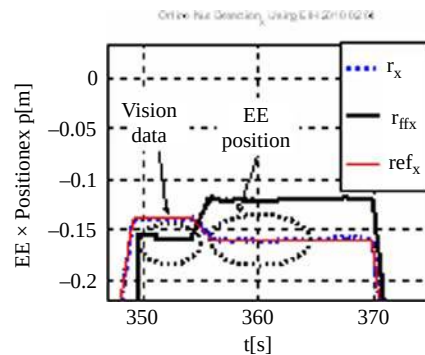
Task 2: vision for the screwing task

Next, we discuss the alignment of the nut and bolt for the screwing pose, by using vision, as shown in Figure 19. From Figure 20(a) and (b), we observed that the y - z position of the

Figure 16 x-position for nut grasping



(a) Position experiment and ETH x data



(b) Position experiment and EIH x data

Table II Repeatability and random location test (%)

Nut size	Repeat	Random
M8	100	100
M4	100	80
M2	90	80

Table III Collision avoidance test for nuts (%)

Sextant	Odd	Even	Total
Trials	5	5	10
Success	100	100	100

Figure 17 Target and obstacle detection

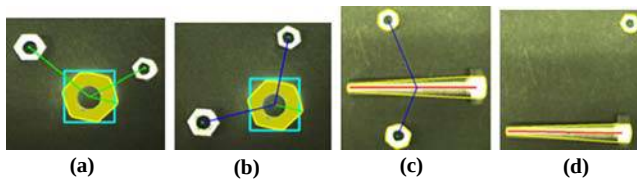
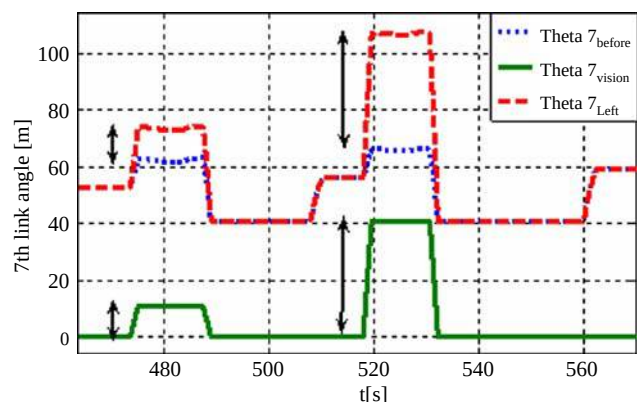


Figure 18 Avoidance angle



left EE (Y_{Exp}, Z_{Exp}) at 807 s was aligned to the position calculated by vision, denoted by (Y_{ff}, Z_{ff}), at 805 s. In addition, we can determine the efficiency of vision by measuring the force at each of the fingers of the left hand that holds the nut when it makes contact with the bolt. We assume that if the nut makes full contact with the surface of the bolt, which is not a desirable result, then the contact force in the z -direction (shown by white arrows in Figure 19(right)) could exceed 1 N (positive and negative signs). In a screwing test with disturbed positions applied to the right arm, the forces measured from the three fingers of the left hand are shown in Figure 21. The green outlined regions indicate the moment of contact based on state values that indicate the robot pose during the experiment. The measured forces ranging from -0.5 to 0.45 N are small. This proves that the center of the nut fits quite well with the bolt. Here, the sharp spikes observed at 248 s in all three graphs are the measured forces at the end of screwing task. These spikes resulted from the forward force caused by pushing the bolt towards the nut to tighten the contact, while rotating the hand between 230 and 248 s, before the hand releases the hold after 249 s.

From the ten trials, the success rate without the integration with the force sensor was only 15 percent. However, with a good integration between vision and force sensors in another ten trials, the autonomous check and re-correct strategy by the robot worked successfully, and increased the success rate to 60 percent. The screen shots of the screwing steps are shown in Figure 22, and numbered from 1 to 9. The recognition of the bolt by vision was initially done in the first picture, which brings the left arm closer to the bolt, as in the second and third pictures. After contact, the position corrections based on force from the left hand were implemented as shown in fourth and fifth pictures, followed by the screwing rotations in the sixth and seventh pictures. From the failed attempts, we observed that the nut and bolt slip easily from the fingers during screwing, for several reasons. First, the current vision data are still insufficient for determining the object posture, which is critical for screwing. Second, the force produced by the nut when it makes contact with the bolt has changed the original alignment of both the nut and bolt. Thus, to tackle these problems in the future, we need to calculate the 6-DOF posture of the targets, and then apply force control, not only to the left hand, but also to the right hand in order to jointly compensate for the force applied on both targets.

Figure 19 Screwing pose

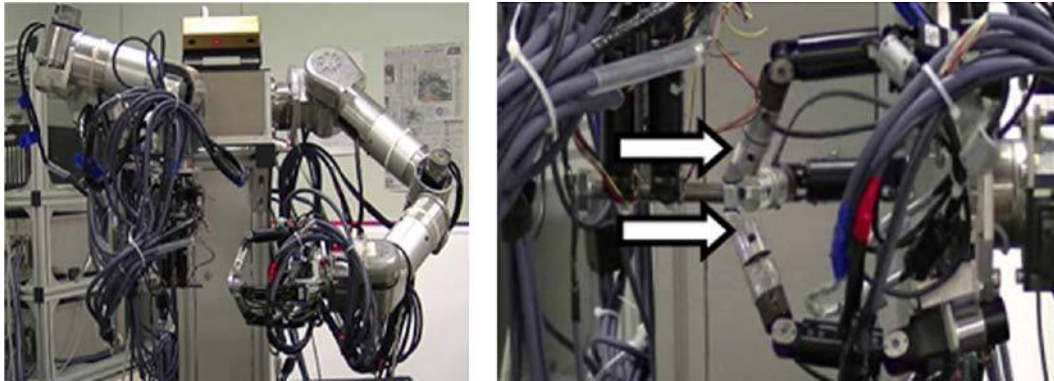


Figure 20 Aligning EE by vision

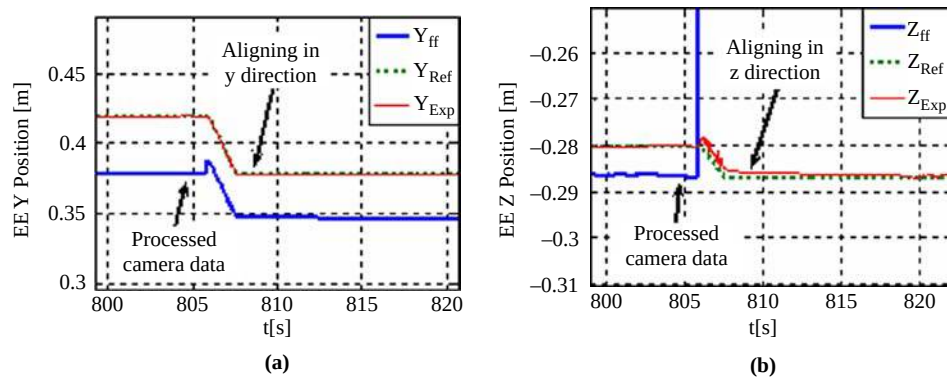


Figure 21 Z-force data

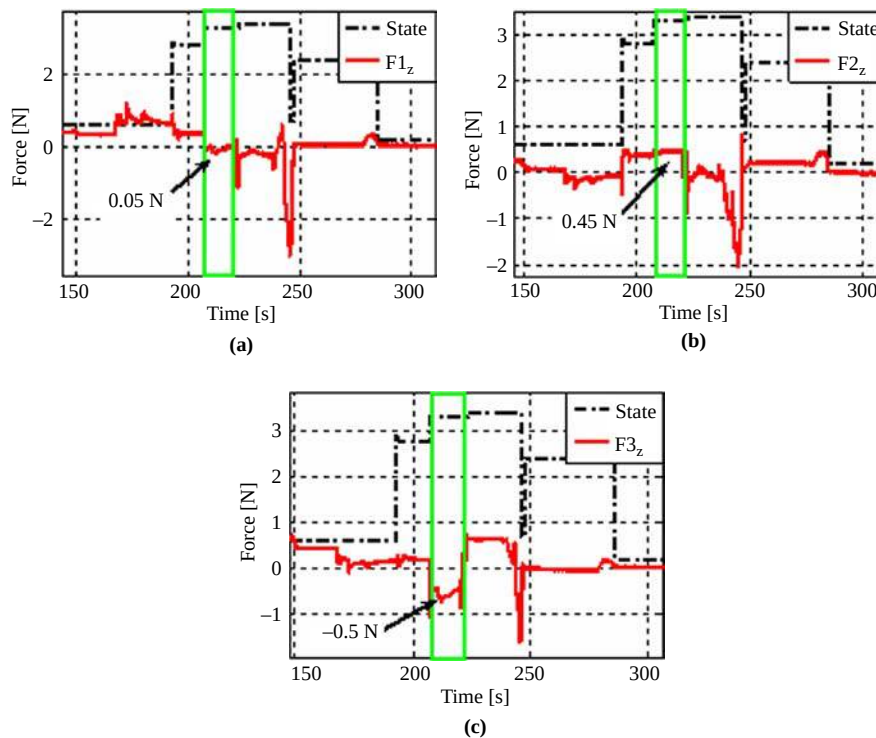
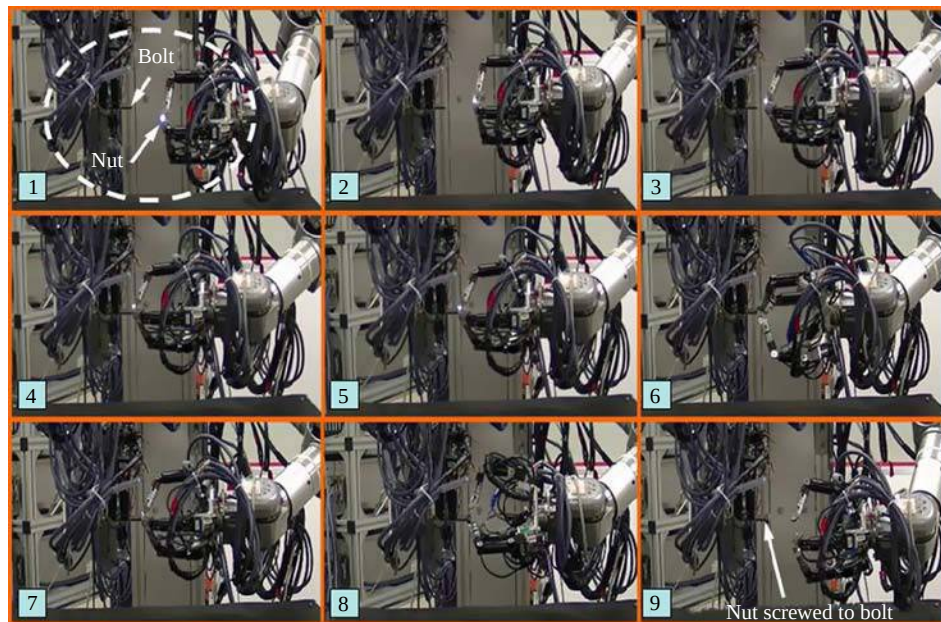


Figure 22 Image sequences of screwing task



Conclusion

In this study, we proposed an application of a dual-arm robot for assembly purposes. We successfully applied a vision algorithm for recognizing small nuts and bolts in an assembly task in which a robot screwed together a nut and a bolt using only three-fingered hands. Real-time experiment results showed that the robot can identify, grasp targets and avoid collisions autonomously using the information processed by vision. We have planned strategies that will enable the robot to control the trajectory of its two arms to accomplish the assigned tasks in real-time autonomous operations. It is also clear that the integration of vision and force sensors could practically improve and contribute to accomplishing complicated tasks involving small objects. This opens new opportunities to improve the capability of dual-arm robots for more challenging tasks in the near future. The calculation of the 6-DOF posture of small targets using vision would be the next step in making the multi-link arm robot be able to change its position and pose according to the complex posture of targets in unstructured conditions.

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